BEFORE YOU READ

One symbol, two meanings

The integral sign is used for two different objects. Whether limits of integration are written decides which one is meant.

The integral

2 types

Definite

$\int_a^b f(x)\,dx$

Result

a number

Means

area or total change

Indefinite

$\int f(x)\,dx = F(x) + C$

Result

a family of antiderivatives

TAKEAWAY

Limits mean a number

With limits the integral is a number; without them it is a family of functions, written with $+C$.